Domain Expertise & Course Alignment Summary

Project Title: TrustRAG: Hallucination Control and Evidence Attribution in Offline RAG

Course Name: RAG with Python: Build Chatbots That Talk to Your Data	**Platform:** Udemy	**Duration:** 10 Hours 34 Minutes
Instructor: Python & AI Academy / Udemy	**Requirement Met:** 10–20 Hours Mandate	**Certificate Retained:** Yes (Verified)
Course URL: https://www.udemy.com/course/rag-with-python-build-chatbots-that-talk-to-your-data/		

1. Course Overview & Domain Relevance

This 10-hour 34-minute course addresses core engineering and theoretical prerequisites for developing end-to-end Retrieval-Augmented Generation (RAG) systems in Python. It covers document chunking, vector embeddings, dense vector databases, conversational state management, and RAG evaluation frameworks. Completing this course satisfies the mandatory 10–20 hour domain expertise requirement for the M.Tech Main Project and provides immediate practical alignment with the **TrustRAG** framework.

2. Course Syllabus & Project Alignment Matrix

Udemy Course Module / Topic	Key Skills & Concepts Covered	Direct Application in TrustRAG Project
1. RAG Fundamentals & Architecture	RAG vs fine-tuning, retrieval pipelines, grounding LLM outputs with facts.	Formulates the offline TrustRAG pipeline baseline architecture.
2. Document Processing & Chunking	PyPDF, Recursive Text Splitters, metadata tracking, chunk size optimization.	Processing MS MARCO passages and local PDF/text doc ingestion.
3. Embeddings & Semantic Search	Sentence-Transformers, vector spaces, cosine similarity, dense retrieval.	Sentence-BERT embedding generation and cosine similarity scoring.
4. Vector Databases & Indexing	Vector stores (FAISS / ChromaDB), persistent indexing, top-k retrieval.	Local FAISS vector store indexing for offline query processing.
5. Conversational RAG & Pipelines	LangChain / LlamaIndex orchestration, memory buffer, prompt engineering.	Orchestrating local Ollama LLM queries with retrieved context.
6. RAG Evaluation & Hallucination	Ragas framework, faithfulness metrics, answer relevance, context precision.	Multi-signal XAI verification (DeBERTa NLI, BERTScore, Tri-State).

3. Technical Justification: How the Course Supports TrustRAG

• **Offline Vector Retrieval & Indexing:** Hands-on experience in building local FAISS indexes with Sentence-BERT embeddings. Forms the foundation for TrustRAG's offline vector retrieval on MS MARCO without external APIs.

• **Explainable Hallucination Verification:** Semantic alignment and faithfulness modules informed TrustRAG's 3-signal XAI engine combining DeBERTa-v3 NLI entailment, BERTScore, and SBERT cosine similarity.

• **Response Suppression Controller:** Studying RAG failure modes guided TrustRAG's Tri-State sentence classification (Supported, Weakly Supported, Hallucinated) and automated response suppression when thresholds are exceeded.

• **End-to-End Local Execution:** Focus on privacy-preserving, local AI application design aligns directly with TrustRAG's zero-cloud, fully offline deployment requirements for sensitive data environments.

Compliance & Retained Artifacts: The course completion certificate and syllabus details have been retained in project documentation. The technical competencies acquired directly enable the design and execution of the M.Tech Main Project.